Question 1: Differential Equations

Consider a second-order linear ordinary differential equation with constant coefficients.

Let y'' + 4y' + 5y = 0 subject to initial conditions y(0) = 1 and y'(0) = -2.

Part A: Determine the characteristic roots of the auxiliary algebraic equation.

Part B: Write the fundamental set of real-valued solutions using Euler's formula.

Part C: Apply the initial conditions to evaluate the arbitrary integration constants.

Part D: Analyze the long-term asymptotic behavior as t approaches infinity.

Part E: State whether the system is underdamped, critically damped, or overdamped.

Part F: Calculate the quasi-frequency of the decaying oscillatory response curve.

Part G: Sketch the phase portrait trajectory in the (y, y') coordinate state space.

Part H: Compute the Laplace transform of the inhomogeneous system with forcing function f(t) = exp(-t).

Part I: Find the steady-state transfer function poles and zeros.

Part J: Justify why uniqueness holds according to the Picard-Lindelof existence theorem.

Question 2: Linear Algebra and Matrix Theory

Let A be a real symmetric matrix of order 4 with known trace and determinant.

Part A: Prove that all eigenvalues of matrix A are strictly real numbers.

Part B: Demonstrate that eigenvectors corresponding to distinct eigenvalues are mutually orthogonal.

Part C: Construct an orthonormal basis for R^4 using the Gram-Schmidt orthogonalization procedure.

Part D: Compute the spectral decomposition A = Q Lambda Q^T explicitly.

Part E: Evaluate the matrix exponential exp(At) using diagonalization techniques.

Part F: State the Cayley-Hamilton theorem and compute the characteristic polynomial p(lambda).

Part G: Express the inverse of matrix A as a linear combination of lower powers.

Part H: Determine whether the associated quadratic form x^T A x is positive definite.

Part I: Calculate the singular value decomposition (SVD) of the rank-deficient variant.

Part J: Compute the Moore-Penrose pseudoinverse for the minimum-norm least squares solution.

Part K: Derive the condition number kappa(A) under the Euclidean matrix 2-norm.

Question 3: Multivariable Calculus and Field Theory

Let F(x, y, z) = (2xy + z^2) i + (x^2 + 2yz) j + (y^2 + 2xz) k be a three-dimensional vector field.

Part A: Compute the curl of vector field F and determine if F is irrotational everywhere.

Part B: Find a scalar potential function phi(x, y, z) such that F equals the gradient of phi.

Part C: Evaluate the line integral of F along a smooth path from (0,0,0) to (1,2,3).

Part D: State the fundamental theorem of line integrals and verify path independence.

Part E: Compute the divergence of F and find the locations of source or sink behavior.

Part F: Let S be the surface of the closed unit sphere centered at the coordinate origin.

Part G: Apply the Gauss Divergence Theorem to evaluate the net outward flux across S.

Part H: Compute the circulation along the boundary curve C bounding the upper hemisphere.

Part I: Verify Stokes' Theorem by directly calculating both the surface and contour integrals.

Part J: Formulate Green's Theorem as a two-dimensional special case in the xy-plane.

Part K: Determine the differential geometric curvature of the resulting streamline curves.

Question 4: Quantum Mechanics and Wave Equations

Consider a quantum particle of mass m confined within an infinite one-dimensional square well potential.

Let the potential V(x) = 0 for 0 < x < L and V(x) = infinity elsewhere.

Part A: Write down the time-independent Schrodinger wave equation for the interior region.

Part B: Apply Dirichlet boundary conditions psi(0) = 0 and psi(L) = 0 to quantize energy levels.

Part C: Derive the normalized stationary state eigenfunctions psi_n(x).

Part D: Calculate the discrete quantized energy eigenvalues E_n as a function of quantum number n.

Part E: Formulate a general superposition state wave packet at initial time t = 0.

Part F: Determine the time evolution operator and evaluate the expectation value of position <x>.

Part G: Compute the momentum expectation value <p> and verify Ehrenfest's theorem.

Part H: Calculate the uncertainty product Delta x * Delta p and verify Heisenberg's principle.

Part I: Find the probability density current j(x, t) and verify probability conservation.

Part J: Discuss the parity symmetry properties of the eigenfunctions under coordinate inversion.

Part K: Analyze the tunneling transmission coefficient through a finite rectangular barrier.

Question 5: Electrodynamics and Wave Propagation

Consider Maxwell's equations in linear isotropic homogeneous non-conducting dielectric media.

Part A: State the four microscopic Maxwell equations in differential vector form.

Part B: Derive the decoupled electromagnetic wave equations for fields E and B from curl equations.

Part C: Compute the propagation phase velocity v in terms of permittivity epsilon and permeability mu.

Part D: Formulate monochromatic plane wave solutions propagating along the positive z-axis.

Part E: Demonstrate the transverse electromagnetic (TEM) nature of free propagating plane waves.

Part F: Calculate the instantaneous and time-averaged Poynting vector S for energy flux density.

Part G: Determine the electromagnetic momentum density and radiation pressure on an absorbing surface.

Part H: Apply electromagnetic boundary conditions across a planar interface between two dielectric media.

Part I: Derive Fresnel's reflection and transmission coefficients for s-polarized (TE) incident waves.

Part J: Derive Fresnel's equations for p-polarized (TM) waves and compute Brewster's polarizing angle.

Part K: Determine the skin depth delta for wave penetration into a conducting metallic waveguide.